MULTI-NODE INFANT CRY DETECTION SYSTEM

Complete Master Hardware Component Specification & Engineering Datasheet

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Target: 3× Cot Nodes + 1× Central Hub + 1× Caregiver Wearable

Capacitor Code Clarification: A **100 nF** (0.1 µF) capacitor is stamped **104** (10 × 10^4 pF = 100,000 pF = 100 nF = 0.1 µF). A capacitor stamped **103** is only 10 nF (0.01 µF), which is 10 times too small. Always use **104** for power decoupling and ADC noise filtering.

1. Subsystem A: Cot Sensing Listening Nodes (3 Cot Units Total)

QUANTITY: 3 IDENTICAL NODES

ID	Component & Make	Manufacturer / Part No.	Electrical Ratings & Specifications	Marking & Package	Circuit Pinout / Role
A1	**Core MCU Node** ESP32-C3 SuperMini	Espressif Systems *ESP32-C3FH4 / FN4*	32-bit RISC-V @ 160 MHz, 400 KB SRAM, 4 MB embedded Flash, 2.4 GHz Wi-Fi / ESP-NOW protocol	Castellated 16-pin dual row (22.5 × 18 mm), Type-C USB	Audio DMA capture & ESP-NOW telemetry sender
A2	**Digital MEMS Microphone** INMP441 I2S Breakout	TDK InvenSense *INMP441ACEZ-R0*	24-bit I2S digital output, 61 dBA SNR, -26 dBFS sensitivity, 60 Hz – 15 kHz, 1.4 mA current draw	6-pin breakout board with bottom acoustic port	`SCK: IO5` `WS: IO6` `SD: IO7` `L/R: GND` `VDD: 3.3V`
A3	**Battery Voltage Divider Resistors**	Generic Metal Film *MF0204 / 0.25W 1%*	100 kΩ resistance, 0.25W (1/4W) rating, ±1% precision tolerance. Form 2:1 ratio for 1S Li-Po	Axial through-hole (4-band): BRN BLK YEL GLD	VBAT → 100 kΩ → `GPIO 1` → 100 kΩ → GND (Vadc = Vbat / 2)
A4	**ADC Noise Filter Capacitor**	Generic MLCC / Ceramic Disc *50V X7R / Y5V*	100 nF (0.1 µF), 50V rated, ±10% tolerance. Low-pass filter for stable ADC readings	Yellow radial bead Stamp: `104` (or 104K / 104Z)	In parallel with lower 100 kΩ resistor between `GPIO 1` and `GND`
A5	**Mic Decoupling Capacitor**	Generic Ceramic Disc *50V Ceramic*	100 nF (0.1 µF), 50V rated. Absorbs digital switching noise from microphone power	Yellow/orange disc Stamp: `104`	Directly across INMP441 `VDD (3.3V)` and `GND`
A6	**Battery Status LED**	Generic 3mm or 5mm *Diffused Red*	Vf = 1.9V – 2.1V, If = 10 mA, 625 nm. High: Solid, Med: Slow Blink, Low: Fast Blink	Longer lead = Anode (+) Flat rim = Cathode (−)	`GPIO 4` → 330 Ω → Anode → Cathode → GND (Active-HIGH)
A7	**ESP-NOW Link Status LED**	Generic 3mm or 5mm *Diffused Blue*	Vf = 3.0V – 3.2V, If = 10 mA, 465 nm. Solid: Connected to Hub, Blink: Searching	Longer lead = Anode (+) Flat rim = Cathode (−)	3.3V → 330 Ω → Anode → Cathode → `GPIO 9` (Active-LOW)
A8	**LED Current-Limiting Resistors**	Generic Carbon / Metal Film *0.25W 5%*	330 Ω resistance, 0.25W rating, ±5% tolerance. Limits LED current safely to ~4–8 mA	Axial through-hole (4-band): ORG ORG BRN GLD	In series with Red LED (GPIO 4) and Blue LED (GPIO 9)
A9	**Li-Ion Charging Module**	TP4056 USB-C Board *TP4056 + DW01A + 8205A*	1S 3.7V constant-current/constant-voltage charger, 1A charge rate, 2.5V cut-off protection	Red/Blue SMD status LEDs, Type-C input socket	BAT+ / BAT− to Li-Po cell; OUT+ / OUT− to power circuit
A10	**DC-DC Step-Up Booster**	MT3608 Micro Boost *MT3608 2A*	Input 2.8V – 4.2V, Output adjusted to 5.0V, 2A peak, 1.2 MHz switching frequency	Solder pad terminals with multiturn trimpot	Steps Li-Po 3.7V up to 5.0V for ESP32 5V/VBUS pin
A11	**RF Power Buffer Capacitor**	Radial Aluminum Electrolytic *47 µF – 100 µF, 25V*	47 µF to 100 µF, 25V rating, 105°C rated. Absorbs sudden 120 mA ESP-NOW RF transmit spikes	Radial can; light vertical stripe marks Negative (−) lead	Across 5V boost converter output rail (5V to GND)

ID	Component & Make	Manufacturer / Part No.	Electrical Ratings & Specifications	Marking & Package	Circuit Pinout / Role
A12	**1S Li-Po Battery Cell**	Rechargeable Li-Po Pouch *Model 103450 / 803450*	1S 3.7V nominal, 2000 mAh capacity, integrated PCM protection board against short-circuits	Flat pouch with pre-crimped red/black lead wires	Connects to TP4056 B+ and B− terminals

Cot Node Strapping Pin Note: GPIO 9 controls the ESP32-C3 boot mode. Our Blue LED design (3.3V → 330 Ω → Anode → Cathode → GPIO 9) naturally pulls the pin HIGH during power-up, ensuring normal SPI flash boot. Do not short GPIO 9 directly to ground on power-on or the chip will enter ROM download mode.

2. Subsystem B: Central Intelligence Hub (1 Unit)

QUANTITY: 1 CENTRAL UNIT

ID	Component & Make	Manufacturer / Part No.	Electrical Ratings & Specifications	Marking & Package	Circuit Pinout / Role
B1	**Central Edge-AI Hub MCU**	Espressif Systems *ESP32-S3-DevKitC-1 N16R8*	Dual-Core Xtensa LX7 @ 240 MHz, 512 KB SRAM, 16 MB Flash, **8 MB Octal PSRAM**. Vector DSP instructions for Log-Mel spectrograms & TinyCryNet inference	44-pin dual-row dev board with dual Type-C ports	ESP-NOW multi-cot receiver, audio reassembly in PSRAM, TFLite Micro inference, BLE alerting server
B2	**2.4" SPI Color TFT Display**	ILI9341 LCD Module *ILI9341 240×320 SPI*	2.4-inch 240×320 pixel, 65k RGB colors, SPI interface (SPI2 hardware bus), onboard 3.3V LDO regulator	14-pin single-row header (2.54 mm pitch)	`CS: IO10` `MOSI: IO11` `SCK: IO12` `DC: IO13` `RST: IO14` `VCC: 5V`
B3	**Backlight Ballast Resistor**	Metal Film Resistor *0.25W 5%*	100 Ω resistance, 0.25W rating. Limits display backlight current to ~20 mA	Axial through-hole (4-band): BRN BLK BRN GLD	Placed between 3.3V rail and TFT `LED/BLK` pin
B4	**DC-DC Step-Up Booster**	MT3608 Micro Boost *MT3608 2A*	Input 3.0V – 4.2V, Output adjusted to 5.0V, 2A peak current capability	Solder terminals with multiturn trimpot	Powers S3 5V/VIN pin and TFT display VCC rail
B5	**High-Capacity Li-Po / Li-Ion**	Rechargeable Lithium *18650 or 3.7V Pouch*	1S 3.7V nominal, 3000 mAh capacity, continuous current rating > 2A with PCM	Single-cell battery with 2-pin JST-PH lead	Primary power reservoir for central hub
B6	**Bulk Decoupling Capacitor**	Aluminum Electrolytic *100 µF, 25V*	100 µF capacitance, 25V rating, low ESR. Stabilizes 5V boost converter output	Radial can; stripe marks Negative (−) lead	Placed across 5V boost converter output rail
B7	**High-Freq Decoupling Cap**	Ceramic Disc / MLCC *50V Ceramic*	100 nF (0.1 µF), 50V rated. Suppresses high-frequency ripple on MCU power bus	Yellow disc Stamp: `104`	Placed across ESP32-S3 3.3V and GND pins

CRITICAL ESP32-S3 OCTAL PSRAM PROTECTION: On the ESP32-S3-WROOM-1 N16R8, **GPIO 33, 34, 35, 36, and 37 are internally dedicated to the 8 MB Octal PSRAM bus**. Wiring anything to these pins will crash the flash/PSRAM cache memory controller. Our TFT display pinout strictly uses **GPIO 10, 11, 12, 13, and 14**, completely protecting the PSRAM bus.

3. Subsystem C: Caregiver Wearable Alert Wristband (1 Unit)

QUANTITY: 1 WEARABLE UNIT

ID	Component & Make	Manufacturer / Part No.	Electrical Ratings & Specifications	Marking & Package	Circuit Pinout / Role
C1	**Wearable MCU Core** ESP32-C3 SuperMini	Espressif Systems *ESP32-C3FH4*	Single-core 32-bit RISC-V @ 160 MHz, 4 MB Flash, BLE 5 Client mode for GATT cry notifications	Ultra-compact castellated 16-pin board (22.5 × 18 mm)	Receives BLE GATT alerts from S3 hub, drives OLED & haptic motor
C2	**0.96" Monochrome OLED**	Solomon Systech SSD1306 *0.96" I2C OLED (0x3C)*	128×64 pixels, I2C interface, 3.3V operating voltage, ultra-low operating current (~15 mA)	4-pin header (VCC, GND, SCL, SDA)	`SDA: IO4` `SCL: IO5` `VCC: 3.3V` `GND: GND`
C3	**Haptic Motor Driver MOSFET**	Alpha & Omega / Inf. (IR) *AO3400A or IRLML2502TR*	Logic-Level N-Channel MOSFET: Vds = 30V, Id = 4.2A – 5.7A, Vgs(th) ≤ 1.2V, Rds(on) < 35 mΩ. Fully turns on at 3.3V logic!	SOT-23 package (or pre-mounted adapter breakout)	`Pin 1: Gate (IO3 via 100 Ω)` `Pin 2: Source (GND)` `Pin 3: Drain (Motor −)`

ID	Component & Make	Manufacturer / Part No.	Electrical Ratings & Specifications	Marking & Package	Circuit Pinout / Role
C4	**Silent Haptic ERM Motor**	Coin Vibration Motor *Model 1027 / 1030*	10 mm flat coin ERM motor, 2.5V – 3.7V DC, nominal current 60 mA – 80 mA, ~10,000 RPM vibration frequency	Self-adhesive coin disc with Red (+) and Blue/Black (–) leads	Red lead to 3.3V rail; Blue/Black lead to MOSFET Drain
C5	**Flyback Snubber Diode**	Diodes Inc. / Vishay *1N5819 Schottky*	Schottky Barrier Diode: Vrrm = 40V, If = 1.0A, low Vf ≈ 0.35V, ultra-fast recovery (< 10 ns). Snubs motor inductive kickback!	DO-41 axial glass/plastic case. **Silver Band = Cathode (–)**	Directly across motor leads: **Silver Band (Cathode) → 3.3V** Black Body (Anode) → MOSFET Drain
C6	**Gate Ringing Resistor**	Carbon / Metal Film *0.25W 5%*	100 Ω resistance, 0.25W rating. Suppresses high-frequency ringing and gate current spikes	Axial through-hole (4-band): BRN BLK BRN GLD	Between `GPIO 3` and MOSFET Gate terminal
C7	**Gate Pulldown Resistor**	Carbon / Metal Film *0.25W 5%*	10 kΩ resistance, 0.25W rating. Guarantees motor stays OFF during boot and floating states	Axial through-hole (4-band): BRN BLK ORG GLD	Between MOSFET Gate terminal and `GND`
C8	**Compact Wearable Li-Po**	Li-Po Pouch Cell *Model 502030 / 602030*	1S 3.7V nominal, 500 mAh capacity, ultra-lightweight pouch (~10 g, 30 × 20 × 5 mm)	Pouch with pre-crimped leads and micro-PCM	Power source for wearable wristband enclosure

4. Subsystem D: Bench Prototyping Hardware, Interconnects & Consumables

WORKSHOP ESSENTIALS

ID	Item & Description	Specification / Model	Engineering Purpose & Notes	Total Quantity
D1	**Solderless Breadboards (Full)**	MB-102 (830 tie points)	Central Hub assembly (ESP32-S3 + ILI9341 display + power modules)	1 Board
D2	**Solderless Breadboards (Half)**	400 tie points	Cot Listening Nodes 1, 2, and 3 bench testing	3 Boards
D3	**Mini Breadboard / Perfboard**	170 tie points / 5 × 7 cm PCB	Caregiver Wearable Wristband prototype layout	1 Board
D4	**Male-to-Male Jumper Wires**	65-piece assorted lengths	Breadboard point-to-point power and signal wiring	2 Packs
D5	**Male-to-Female Dupont Wires**	40-pin 20 cm ribbon cable	Direct connection from breadboards to INMP441, ILI9341, and SSD1306 headers	2 Packs
D6	**High-Speed USB-C Data Cables**	1.0 m shielded USB-A to USB-C	Must support data sync (D+ / D− lines intact), not power-only charging cables	3 Cables
D7	**2-Pin JST-PH 2.0 mm Pigtails**	Male & Female pre-crimped leads	Modular battery quick-disconnects for TP4056 charging boards	5 Pairs
D8	**Rosin-Core Solder Wire**	63/37 Sn/Pb (or Sn-Cu Lead-Free), 0.8 mm	Soldering header pins to ESP32-C3 SuperMini, INMP441, and flyback diode	1 Spool (50g)
D9	**Heat-Shrink Tubing Assortment**	1.5 mm, 2.5 mm, 4.0 mm polyolefin	Insulating diode leads, motor joints, and battery terminals safely	1 Assortment Kit

5. Master Hardware Identification Quick-Reference

BENCH REFERENCE

Component Type	Value / Spec	Physical Stamp / Marking	Color Bands / Polarity Rule
Decoupling / Filter Capacitor	100 nF (0.1 µF, 50V)	`104` (or 104K, 104Z)	Non-polar ceramic disc (either direction)
Bulk Buffer Capacitor	47 µF – 100 µF (25V)	Values printed on can (e.g. 100µF 25V)	White/light stripe marks Negative (−) lead (shorter leg)
Battery Divider Resistor	100 kΩ (0.25W 1%)	—	BRN BLK YEL GLD
LED Current-Limiting Resistor	330 Ω (0.25W 5%)	—	ORG ORG BRN GLD
Gate Ringing / Backlight Resistor	100 Ω (0.25W 5%)	—	BRN BLK BRN GLD
Gate Pulldown Resistor	10 kΩ (0.25W 5%)	—	BRN BLK ORG GLD
Flyback Snubber Diode	1N5819 (40V 1A Schottky)	`1N5819` printed on body	**Silver band marks Cathode (−)** (connect to 3.3V)